

3 Beginner AI Projects · Logical Flow

· Project Overview

Three beginner-friendly AI projects: a ReAct Agent, an AI Resume Critiquer, and an Image Classifier.

- Project 1: ReAct Agent (Tool-Calling AI Assistant)

```

    ReAct Agent Flow
    .....
    User Input (CLI)
        .
        .
    LangGraph ReAct Agent
        .
        .
        .....
        .   Agent Reasoning   .
        .   (Think · Act)     .
        .....
            .
            .....
                .
                .
    @calculator   @say_hello
    (math)         (greeting)
        .           .
        .....
            .
            .
    Tool Execution Result
        .
        .
    Response Generation
        .
        .
    Display to User ... Loop until "quit"

```

- Project 2: AI Resume Critiquer

```

.....
·                               Resume Critiquer Flow
.....
·
·   Streamlit Web UI
·       ·
·       ·
·   File Upload (PDF / TXT)

```

```

.
.
.      ..... PDF? ... PyPDF2 Text Extraction
.      .
.      ..... TXT? ... Direct Text Decode
.      .
.      .....
.      .
.      Extracted Text
.      .
.      .
.      User Specifies Target Role
.      .
.      .
.      OpenAI API Analysis
.      (System Prompt + Resume)
.      .
.      .
.      Structured Feedback Display
.      . Strengths
.      . Weaknesses
.      . Suggestions
.
.
. ....

```

· Project 3: Image Classifier

```

. ....
.      Image Classifier Flow
. ....
.
.      Streamlit Web UI
.      .
.      .
.      Image Upload (JPG/PNG)
.      .
.      .
.      Preprocessing
.      ... Resize to 224x224
.      ... Normalize pixel values
.      ... Prepare input tensor
.      .
.      .
.      MobileNetV2 (Pre-trained)
.      ... ImageNet weights
.      .
.      .
.      Model Inference
.      .
.      .
.      decode_predictions()
.      .
.      .

```

```
.  Display Top-3 Results .
.  ... Class Name 1 . Confidence % .
.  ... Class Name 2 . Confidence % .
.  ... Class Name 3 . Confidence % .
.
.....
```